



BDS Series

AHU Double Skin



Tasaki

AHU Double Skin

COMPANY PROFILE

Bitwise group of companies were established in 1988 in samutprakarn province by a determined team of engineers who desired to manufacture international standard air conditioner to cater to the needs of OEM and ODM customers.

Some years after very well success in Thailand, the company expanded its factory line to China. These companies are known as Shanghai Bitwise Electric Motor & Appliances, Yongwise (Shanghai) Molding Co., Ltd. And Shanghai Bitwise Freezing Equipment Co., Ltd.

Today, Bitwise Group has its production area of 120,000 square meters, the R&D team, highly qualified and experienced engineers, technicians, over 1,000 workers. The company has successfully developed a complete range of air conditioner products from 1,000 to 2,000,000 Btu/Hr, evaporator and condenser coils for air conditioners, as well as for the frozen food industry, evaporative air cooler and water heater, IQF freezer etc.

Apart from air conditioner industry, Bitwise Group also manufactures various types of different products such as resin packed motors, PCBA, LED bulbs, mould manufacturing and plastic injections, public telephone booths entrusted by government telecommunication authorities.

With over 20 years experience in the air conditioner industry, through all the international standards it had achieved, Bitwise Group has demonstrated that it will continue to develop further innovation and technologies by its own R&D teams. As a result, it is one of the main ODM players in this field and will continue to constantly grow together with its customers.



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SPECIFICATIONS STANDARD



MODEL	DIMENSION		AIR FLOW	Coil Face Area	Coil Face Velocity
	Width(mm)	Height(mm)	(CFM)	(m2)	(m/s)
25BDS020	980	880	1,900	0.37	2.41
25BDS030	1285	880	2,900	0.56	2.45
25BDS040	1590	880	3,900	0.74	2.47
25BDS060	1590	1185	5,900	1.12	2.49
25BDS080	1590	1490	7,900	1.49	2.50
25BDS100	1895	1490	9,800	1.86	2.49
25BDS120	2200	1490	11,800	2.23	2.49
25BDS150	2200	1795	14,800	2.79	2.50
25BDS180	2200	2100	17,700	3.35	2.49
25BDS210	2505	2100	20,700	3.91	2.50
25BDS240	2810	2100	23,700	4.47	2.50
25BDS280	2810	2405	27,600	5.21	2.50
25BDS320	2810	2710	31,600	5.95	2.50
25BDS360	3115	2710	35,500	6.70	2.50

Dimension Calculation

External Dimension AHU Width : Width

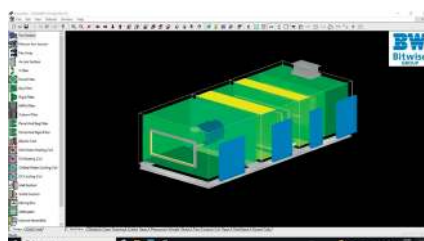
External Dimension AHU Height : Height + Base AHU

Example : Model 25BDS020

External Dimension AHU Width : 980 mm.

External Dimension AHU Height : $880 + 100 = 980$ mm.

COMPUTER PROGRAM



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SPECIFICATIONS STANDARD



MODEL	DIMENSION		AIR FLOW	Coil Face Area	Coil Face Velocity
	Width(mm)	Height(mm)	(CFM)	(m2)	(m/s)
50BDS020	1030	930	1,900	0.37	2.41
50BDS030	1335	930	2,900	0.56	2.45
50BDS040	1640	930	3,900	0.74	2.47
50BDS060	1640	1235	5,900	1.12	2.49
50BDS080	1640	1540	7,900	1.49	2.50
50BDS100	1945	1540	9,800	1.86	2.49
50BDS120	2250	1540	11,800	2.23	2.49
50BDS150	2250	1845	14,800	2.79	2.50
50BDS180	2250	2150	17,700	3.35	2.49
50BDS210	2555	2150	20,700	3.91	2.50
50BDS240	2860	2150	23,700	4.47	2.50
50BDS280	2860	2455	27,600	5.21	2.50
50BDS320	2860	2760	31,600	5.95	2.50
50BDS360	3165	2760	35,500	6.70	2.50
50BDS400	3470	2760	39,500	7.44	2.50
50BDS440	3775	2760	43,400	8.19	2.50
50BDS480	4080	2760	47,400	8.93	2.50

Dimension Calculation

External Dimension AHU Width : Width

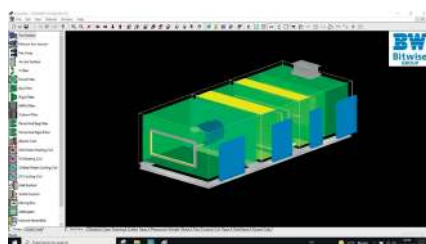
External Dimension AHU Height : Height + Base AHU

Example : Model 50BDS020

External Dimension AHU Width : 1030 mm.

External Dimension AHU Height : $930 + 100 = 1030$ mm.

COMPUTER PROGRAM



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SPECIFICATIONS STANDARD



MODEL	DIMENSION		AIR FLOW	Coil Face Area	Coil Face Velocity
	Width(mm)	Height(mm)	(CFM)	(m2)	(m/s)
60BDS020	1050	950	1,900	0.37	2.14
60BDS030	1355	950	2,900	0.56	2.45
60BDS040	1660	950	3,900	0.74	2.47
60BDS060	1660	1255	5,900	1.12	2.49
60BDS080	1660	1560	7,900	1.49	2.50
60BDS100	1965	1560	9,800	1.86	2.49
60BDS120	2270	1560	11,800	2.23	2.49
60BDS150	2270	1865	14,800	2.79	2.50
60BDS180	2270	2170	17,700	3.35	2.49
60BDS210	2575	2170	20,700	3.91	2.50
60BDS240	2880	2170	23,700	4.47	2.50
60BDS280	2880	2475	27,600	5.21	2.50
60BDS320	2880	2780	31,600	5.95	2.50
60BDS360	3185	2780	35,500	6.70	2.50
60BDS400	3470	2780	39,500	7.44	2.50
60BDS440	3795	2780	43,400	8.19	2.50
60BDS480	4100	2780	47,400	8.93	2.50

Dimension Calculation

External Dimension AHU Width : Width

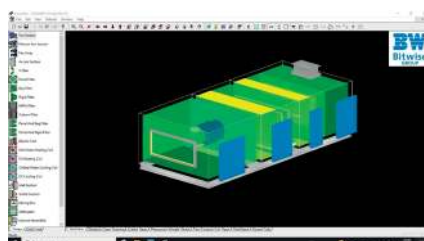
External Dimension AHU Height : Height + Base AHU

Example : Model 60BDS020

External Dimension AHU Width : 1050 mm.

External Dimension AHU Height : 950 + 100 = 1050 mm.

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SPECIFICATIONS STANDARD



MODEL	DIMENSION		AIR FLOW	Coil Face Area	Coil Face Velocity
	Width(mm)	Height(mm)	(CFM)	(m2)	(m/s)
80BDS020	1090	990	1,900	0.37	2.14
80BDS030	1395	990	2,900	0.56	2.45
80BDS040	1700	990	3,900	0.74	2.47
80BDS060	1700	1255	5,900	1.12	2.49
80BDS080	1700	1600	7,900	1.49	2.50
80BDS100	2005	1600	9,800	1.86	2.49
80BDS120	2310	1600	11,800	2.23	2.49
80BDS150	2310	1905	14,800	2.79	2.50
80BDS180	2310	2210	17,700	3.35	2.49
80BDS210	2615	2210	20,700	3.91	2.50
80BDS240	2920	2210	23,700	4.47	2.50
80BDS280	2920	2515	27,600	5.21	2.50
80BDS320	2920	2820	31,600	5.95	2.50
80BDS360	3225	2820	35,500	6.70	2.50
80BDS400	3530	2820	39,500	7.44	2.50
80BDS440	3835	2820	43,400	8.19	2.50
80BDS480	4140	2820	47,400	8.93	2.50

Dimension Calculation

External Dimension AHU Width : Width

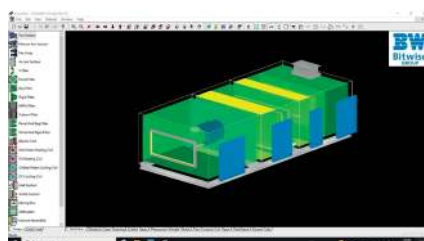
External Dimension AHU Height : Height + Base AHU

Example : Model 80BDS020

External Dimension AHU Width : 1090 mm.

External Dimension AHU Height : 990 + 100 = 1090 mm.

COMPUTER PROGRAM



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MODEL	DIMENSION		AIR FLOW	Coil Face Area	Coil Face Velocity
	Width(mm)	Height(mm)	(CFM)	(m2)	(m/s)
90BDS020	1110	1010	1,900	0.37	2.14
90BDS030	1415	1010	2,900	0.56	2.45
90BDS040	1720	1010	3,900	0.74	2.47
90BDS060	1720	1315	5,900	1.12	2.49
90BDS080	1720	1620	7,900	1.49	2.50
90BDS100	2025	1620	9,800	1.86	2.49
90BDS120	2330	1620	11,800	2.23	2.49
90BDS150	2330	1925	14,800	2.79	2.50
90BDS180	2330	2230	17,700	3.35	2.49
90BDS210	2635	2230	20,700	3.91	2.50
90BDS240	2940	2230	23,700	4.47	2.50
90BDS280	2940	2535	27,600	5.21	2.50
90BDS320	2940	2840	31,600	5.95	2.50
90BDS360	3245	2840	35,500	6.70	2.50
90BDS400	3550	2840	39,500	7.44	2.50
90BDS440	3855	2840	43,400	8.19	2.50
90BDS480	4160	2840	47,400	8.93	2.50

Dimension Calculation

External Dimension AHU Width : Width

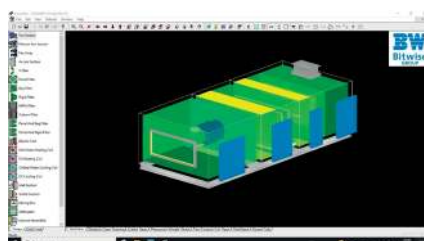
External Dimension AHU Height : Height + Base AHU

Example : Model 90BDS030

External Dimension AHU Width : 1415 mm.

External Dimension AHU Height : $1010 + 100 = 1110$ mm.

COMPUTER PROGRAM



For addition, please refer to technical data Sheet "Door System of Air Handling Units"

DOUBLE SKIN PANEL 50 MM.

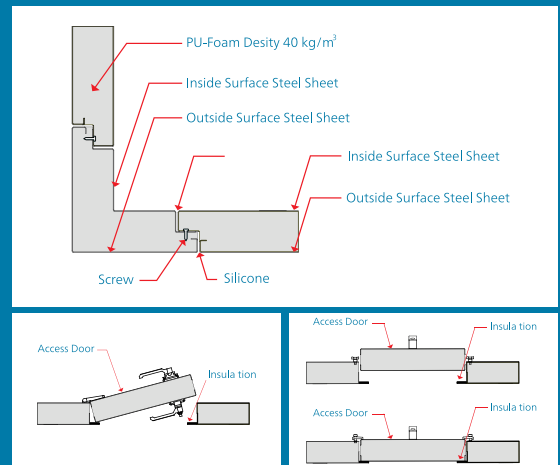
Panel 50 mm.

Standard 50 mm. thick infill panel are of double skin construction from pressure Injected polyurethane foam insulation

- Thermal conductivity (k) = 0.0204 W/m.K (0.0118 Btu/ft.h.°F)

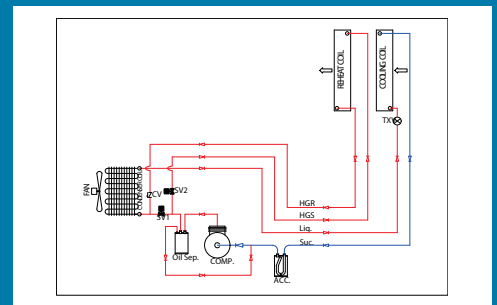
- Heat transfer coefficient of panel 50 mm thickness (k/L) = 0.408 W/m².K (0.0719 Btu/ ft².h.°F)

- Density = 40 kg/m³



HOT GAS RE HEAT

when the compressor is running to reduce the temperature and when the compressor is running to reduce the temperature and temperature to reduce humidity. In this case, the compressor will send refrigerant flow to the reheat coil to compensate for excessive cooling.



FILTER SECTION

Filters are designed for easy removal and renewal of filter cells. and created for use with any type of Pre-filter, Medium filter and Hepa filter Effective diversity in areas of particular importance



Pre Filter



Med Filter



Hepa Filter

COIL SECTION

Bitwise's AHU has its coil constructed with aluminium corrugated fin and copper plain tube 3/8", 1/2" and 5/8". Copper fins and hydrophilic fins are anti-corrosive materials which are optional. The fins are designed purposely for better heat transfer efficiency and moisture carry over limit performance. Capacity, pressure drop and selection procedure are designed in according with ARI standard 410.

The design face velocity is 2.5m/s and for higher face velocity must use eliminator to prevent carry over. If the unit has stack coil design as the coil section that Bitwise AHU provide medium drain-pan between upper coil and lower coil, this prevent water flooding the lower coil.



FAN SECTION

Some application, such as cleanroom need to provide clean air. Plenum fan can support about it. Also, plenum fan contribute to lower overall system pressure drop, thereby reducing energy consumption and can use it with inverter control



MOTOR

Normally, we provide 4 poles, 380V-415V/3Ph/50Hz with IP55 class F to standard models and also we have to option motor as

1. 2 & 6 poles motor
2. Economic efficiency motor.
3. Hi efficiency motor
4. 50Hz motor

There are a few components which are able to provide safety, efficiency and flexibility for the operation of AHU. It includes thermistor, Inverter, disconnect switch and other.

ACCESSORIES



VOLUME DAMPER



WATER PROOF SW.



LAMP



INSPECTION WINDOW





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**SPECIFICATIONS AND INFORMATION IN THIS BROCHURE
ARE SUBJECT TO CHANGE WITHOUT NOTICE**

รายละเอียดดังกล่าวอาจมีการเปลี่ยนแปลงโดยไม่另行通知ให้ทราบล่วงหน้า

ISO 50001 : 2018

ISO 9001 : 2015

ISO 14001 : 2015

ISO 45001 : 2018

ISO/IEC 17025 : 2017



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